
Import data from different sources

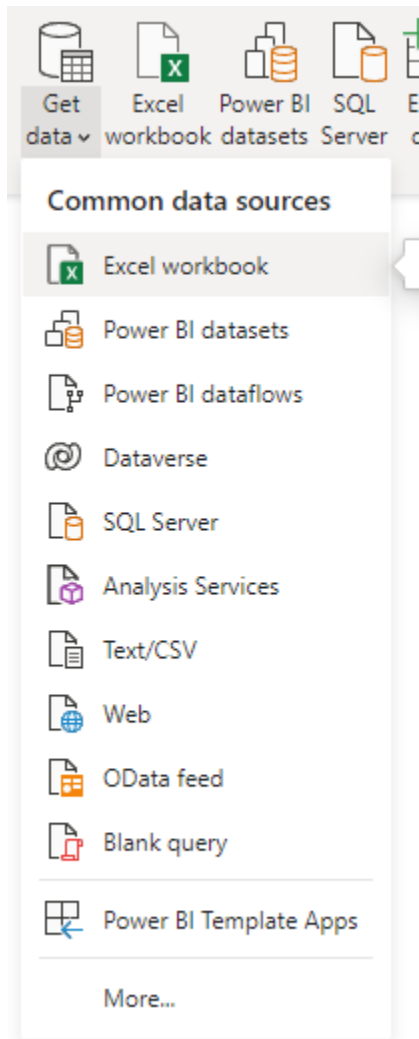
It is common to have data in multiple data sources. For example, you could have two databases, one for product information, and another for sales information. With *Power BI Desktop*, you can combine data from different sources to create interesting, compelling data analyses and visualizations.

In this tutorial, you combine data from two data sources:

- An Excel workbook with product information (you can download this from Blackboard – Products file)
- An OData feed containing orders data (you will use a link to get this file)

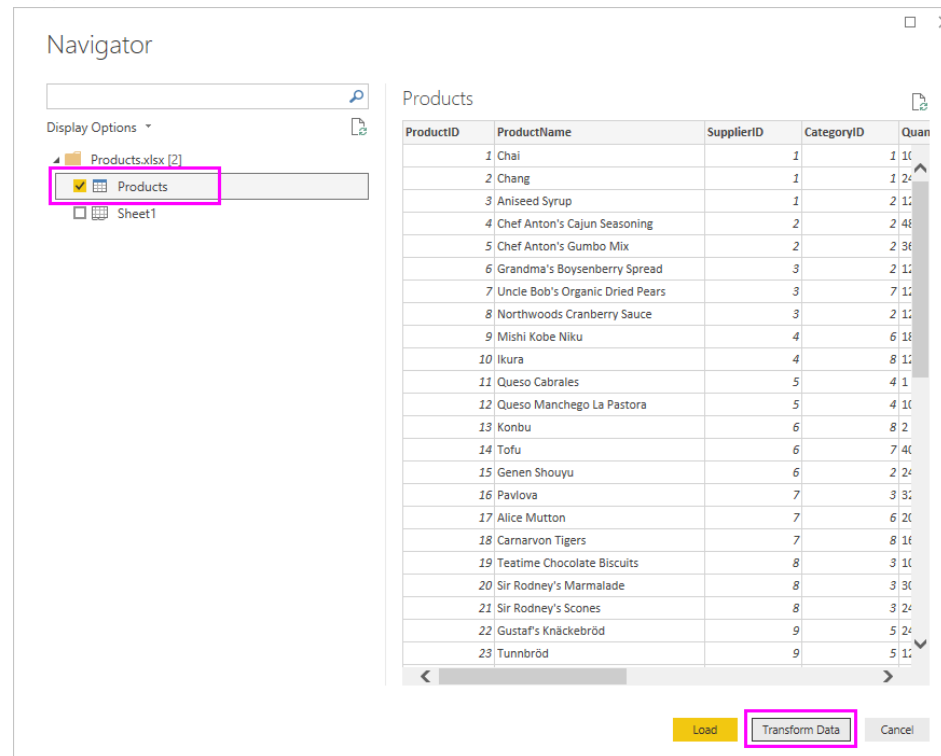
You are going to import each dataset and do transformation and aggregation operations. Then, you will use the two source's data to produce a sales analysis report with interactive visualizations.

Step 1: Import Excel product data

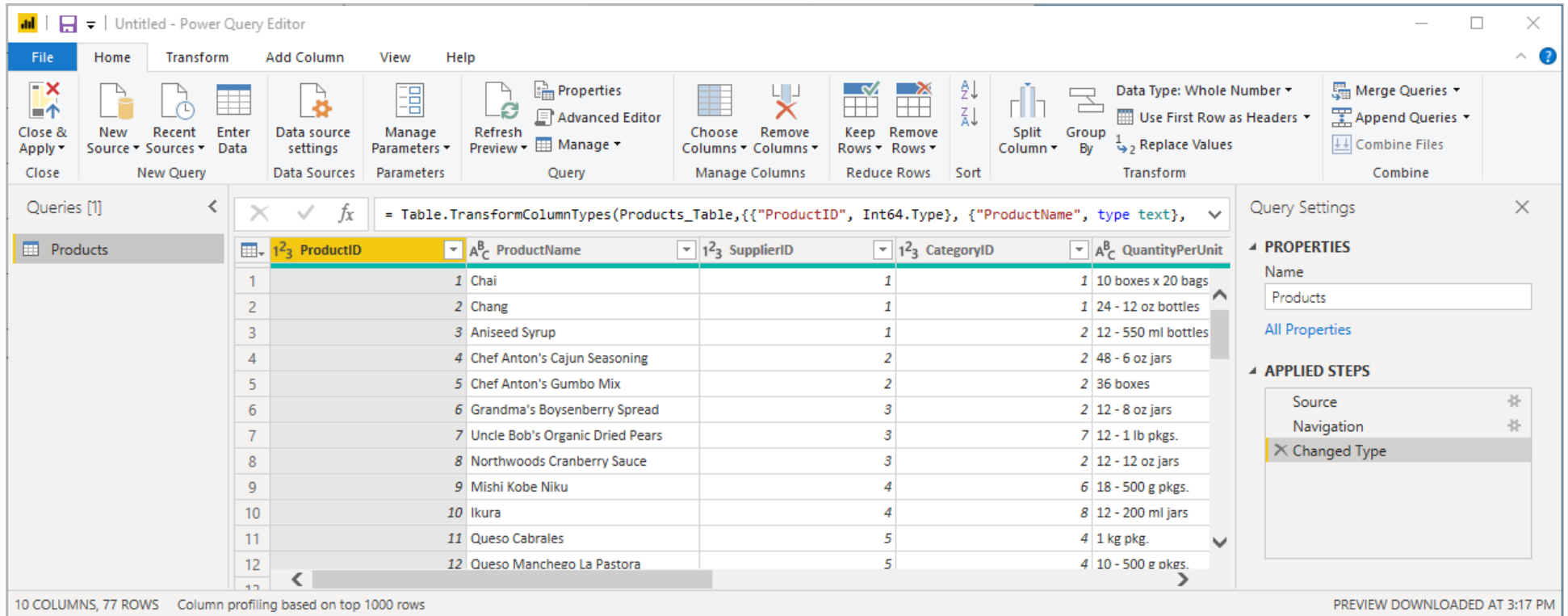


First, let's import product data from the *Products.xlsx* Excel workbook into Power BI Desktop.

1. Download the file *Products.xlsx* from Blackboard and save it in your machine.
2. Select the arrow next to **Get Data** in the Power BI Desktop ribbon's **Home** tab, and then select **Excel** file.
3. In the **Open** dialog box, navigate to and select the **Products.xlsx** file, and then select **Open**.
4. In the **Navigator**, select the **Products** table and then select **Transform Data**.



A table preview opens in the Power Query Editor, where you can apply transformations to clean up the data. We will transform the data in the next sections.



The screenshot shows the Power Query Editor window titled "Untitled - Power Query Editor". The ribbon includes tabs for File, Home, Transform, Add Column, View, and Help. The Transform tab is active, showing various data manipulation options like Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Refresh Preview, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Sort, Split Column, Group By, Data Type, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, and Combine Files.

The main area displays a table with the following columns: ProductID, ProductName, SupplierID, CategoryID, and QuantityPerUnit. The table contains 12 rows of data, including products like Chai, Chang, Aniseed Syrup, and Queso Manchego La Pastora.

The Query Settings pane on the right shows the query name "Products" and the applied steps: Source, Navigation, and Changed Type.

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit
1	Chai	1	1	10 boxes x 20 bags
2	Chang	1	1	24 - 12 oz bottles
3	Aniseed Syrup	1	2	12 - 550 ml bottles
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars
5	Chef Anton's Gumbo Mix	2	2	36 boxes
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.
10	Ikura	4	8	12 - 200 ml jars
11	Queso Cabrales	5	4	1 kg pkg.
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.

10 COLUMNS, 77 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 3:17 PM

Your combined report will use the Excel workbook's **ProductID**, **ProductName**, **QuantityPerUnit**, and **UnitsInStock** columns.

You can remove the other columns.

To do so, select the 4 columns that you want to keep while pressing Ctrl on your keyboard, then right click on the column headers

Query Settings

= Table.TransformColumnTypes(Products_Table,{{"ProductID", Int64.Type}, {"ProductName", type text}, {"SupplierID", Int64.Type}, {"CategoryID", Int64.Type},

	ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice	UnitsInStock	UnitsOnOrder
1	1	Chai	1	1	10 boxes x 20 bags	18		
2	2	Chang	1	1	24 - 12 oz bottles	19		
3	3	Aniseed Syrup	1	2	12 - 550 ml bottles	10		
4	4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22		
5	5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35		
6	6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25		
7	7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30		
8	8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40		
9	9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97		
10	10	Ikura	4	8	12 - 200 ml jars	31		
11	11	Queso Cabrales	5	4	1 kg pkg.	21		
12	12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38		
13	13	Konbu	6	8	2 kg box	6		
14	14	Tofu	6	7	40 - 100 g pkgs.	23.25		
15	15	Genen Shouyu	6	2	24 - 250 ml bottles	15.5		
16	16	Pavlova	7	3	32 - 500 g boxes	17.45		
17	17	Alice Mutton	7	6	20 - 1 kg tins	39		
18	18	Carnarvon Tigers	7	8	16 kg pkg.	62.5		42
19	19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	9.2		25
20	20	Sir Rodney's Marmalade	8	3	30 gift boxes	81		40
21	21	Sir Rodney's Scones	8	3	24 pkgs. x 4 pieces	10		3
22	22	Gustaf's Knäckebröd	9	5	24 - 500 g pkgs.	21		104
23	23	Tunnbröd	9	5	12 - 250 g pkgs.	9		61
24	24	Guaraná Fantástica	10	1	12 - 355 ml cans	4.5		20
25	25	NuNuCa Nuß-Nougat-Creme	11	3	20 - 450 g glasses	14		76
26	26	Gumbär Gummibärchen	11	3	100 - 250 g bags	31.23		15
27	27	Schoggi Schokolade	11	3	100 - 100 g pieces	43.9		49
28	28	Rösste Sauerkraut	12	7	25 - 825 g cans	45.6		26

Copy
Remove Columns
Remove Other Columns
Add Column From Examples...
Remove Duplicates
Remove Errors
Replace Values...
Fill
Change Type
Merge Columns
Group By...
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Move

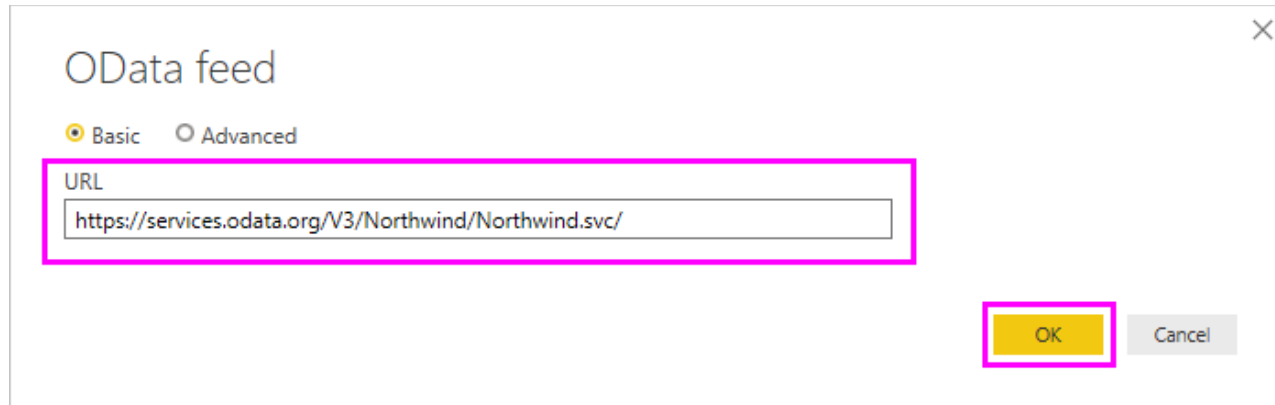
and select **Remove Other Columns**.

Then click on **File -> Close&Apply** to apply your changes.

Step 2: Import OData product data

Next, import the order data from the sample Northwind sales system OData feed.

1. In Power Query Desktop, select **Get Data** and then select **OData feed**. OData means Open Protocol Data, you can think of this as an external source that you can use to retrieve data.
2. In the **OData feed** dialog box, paste the Northwind OData feed URL, <https://services.odata.org/V3/Northwind/Northwind.svc/> Select **OK**. The **Northwind database** is a sample **database** used by Microsoft to demonstrate the features of some of its products, including SQL Server and Microsoft Access. The **database** contains the sales data for **Northwind Traders**, a fictitious specialty foods export-import company (see below).



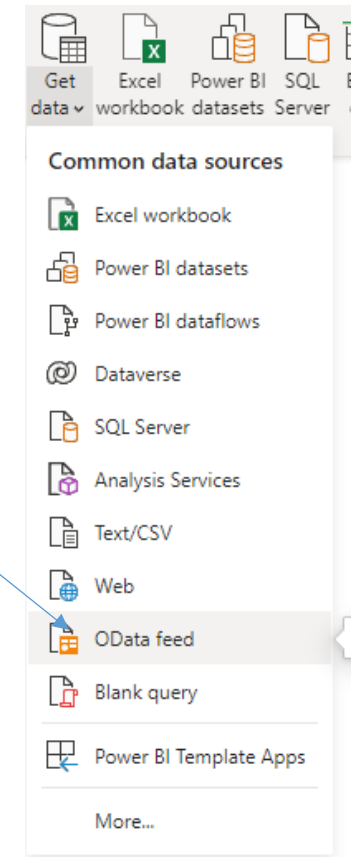
OData feed

☒ Basic ☐ Advanced

URL

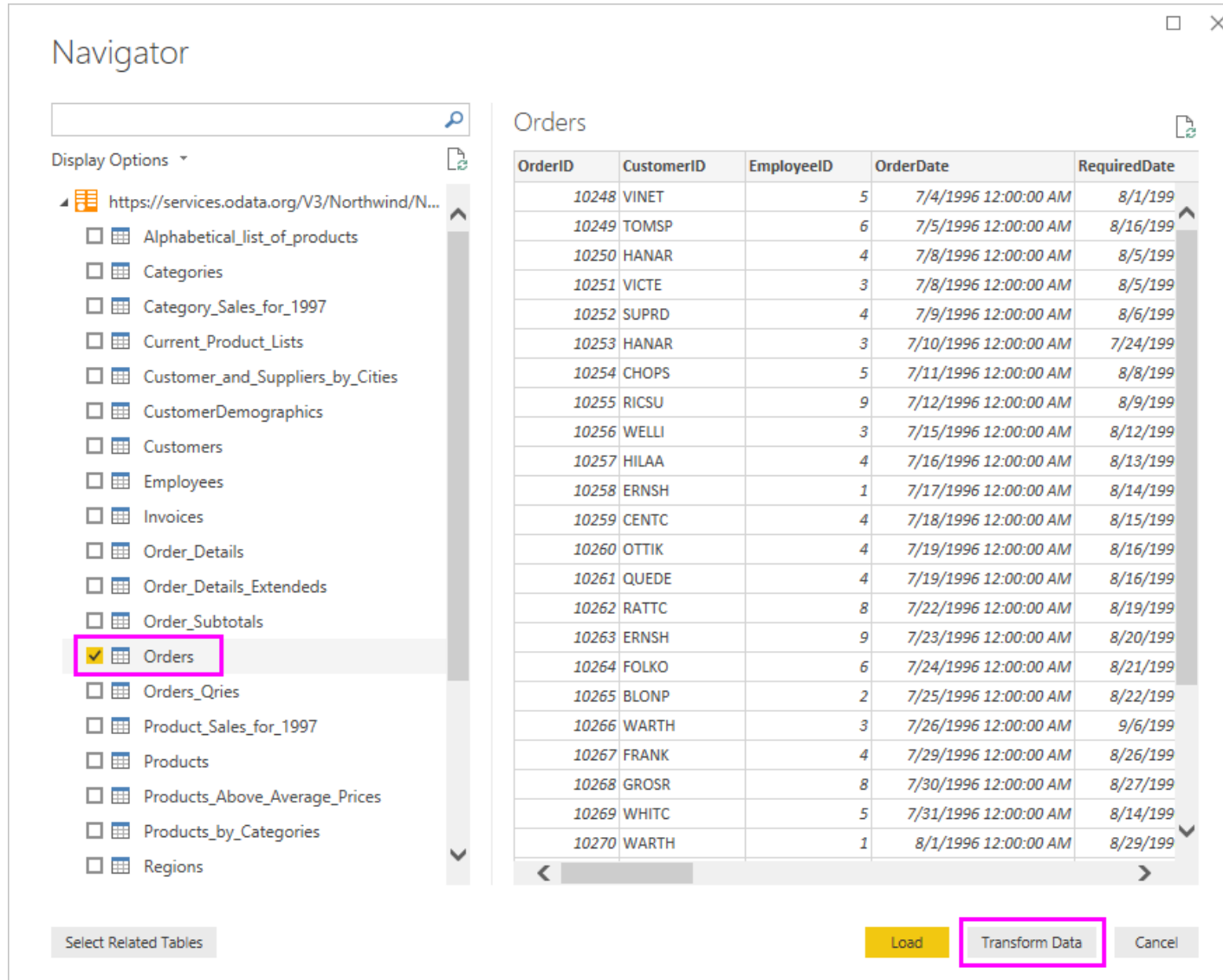
<https://services.odata.org/V3/Northwind/Northwind.svc/>

OK Cancel



Click **Connect** in the new OData Feed window.

3. In **Navigator**, select the **Orders** table, and then select **Transform Data** to load the data into Power Query Editor.



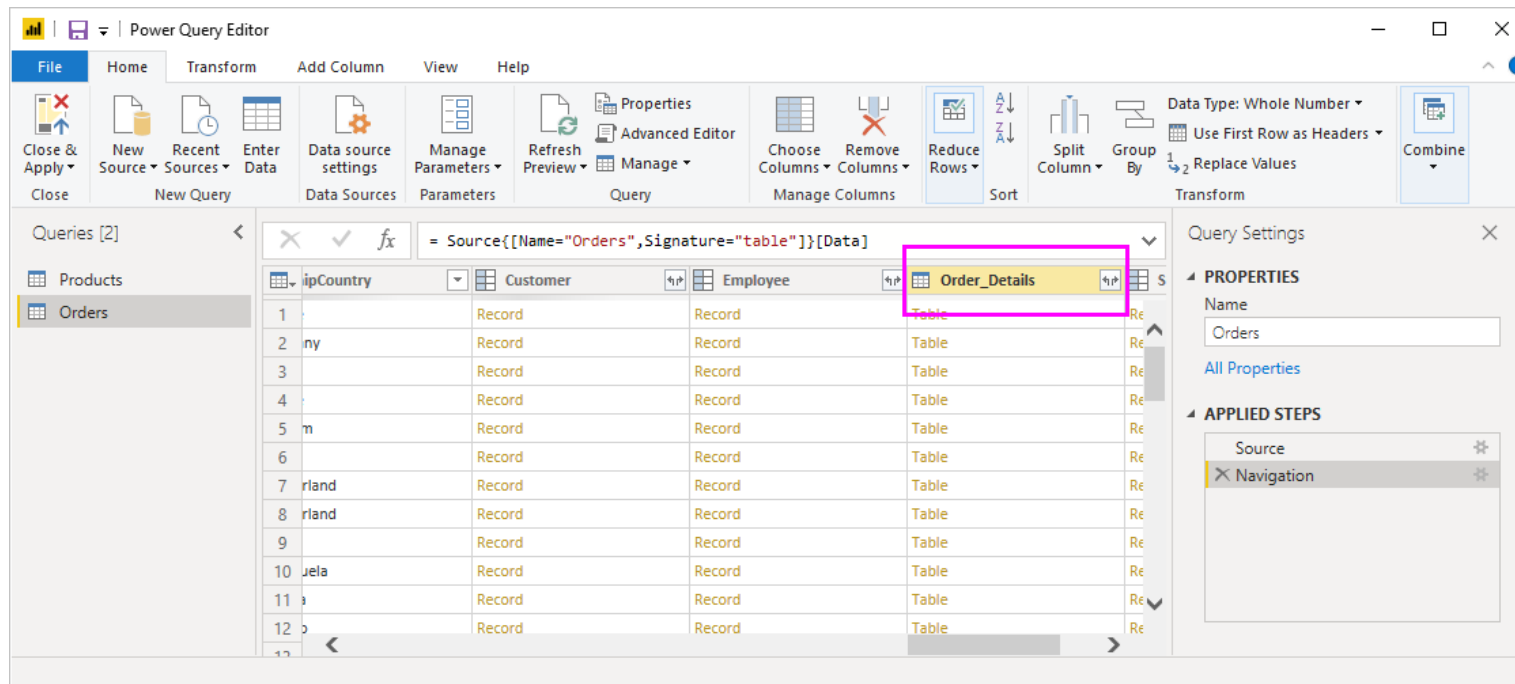
The screenshot shows the Power Query Navigator window. On the left, the 'Navigator' pane displays a list of tables under the URL 'https://services.odata.org/V3/Northwind/N...'. The 'Orders' table is selected, indicated by a checkmark and a pink highlight. On the right, the 'Orders' table preview is shown with columns: OrderID, CustomerID, EmployeeID, OrderDate, and RequiredDate. At the bottom right, the 'Transform Data' button is highlighted with a pink box, alongside 'Load' and 'Cancel' buttons.

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate
10248	VINET	5	7/4/1996 12:00:00 AM	8/1/199
10249	TOMSP	6	7/5/1996 12:00:00 AM	8/16/199
10250	HANAR	4	7/8/1996 12:00:00 AM	8/5/199
10251	VICTE	3	7/8/1996 12:00:00 AM	8/5/199
10252	SUPRD	4	7/9/1996 12:00:00 AM	8/6/199
10253	HANAR	3	7/10/1996 12:00:00 AM	7/24/199
10254	CHOPS	5	7/11/1996 12:00:00 AM	8/8/199
10255	RICSU	9	7/12/1996 12:00:00 AM	8/9/199
10256	WELLI	3	7/15/1996 12:00:00 AM	8/12/199
10257	HILAA	4	7/16/1996 12:00:00 AM	8/13/199
10258	ERNSH	1	7/17/1996 12:00:00 AM	8/14/199
10259	CENTC	4	7/18/1996 12:00:00 AM	8/15/199
10260	OTTIK	4	7/19/1996 12:00:00 AM	8/16/199
10261	QUEDE	4	7/19/1996 12:00:00 AM	8/16/199
10262	RATTC	8	7/22/1996 12:00:00 AM	8/19/199
10263	ERNSH	9	7/23/1996 12:00:00 AM	8/20/199
10264	FOLKO	6	7/24/1996 12:00:00 AM	8/21/199
10265	BLONP	2	7/25/1996 12:00:00 AM	8/22/199
10266	WARTH	3	7/26/1996 12:00:00 AM	9/6/199
10267	FRANK	4	7/29/1996 12:00:00 AM	8/26/199
10268	GROSR	8	7/30/1996 12:00:00 AM	8/27/199
10269	WHITC	5	7/31/1996 12:00:00 AM	8/14/199
10270	WARTH	1	8/1/1996 12:00:00 AM	8/29/199

Expand the order data

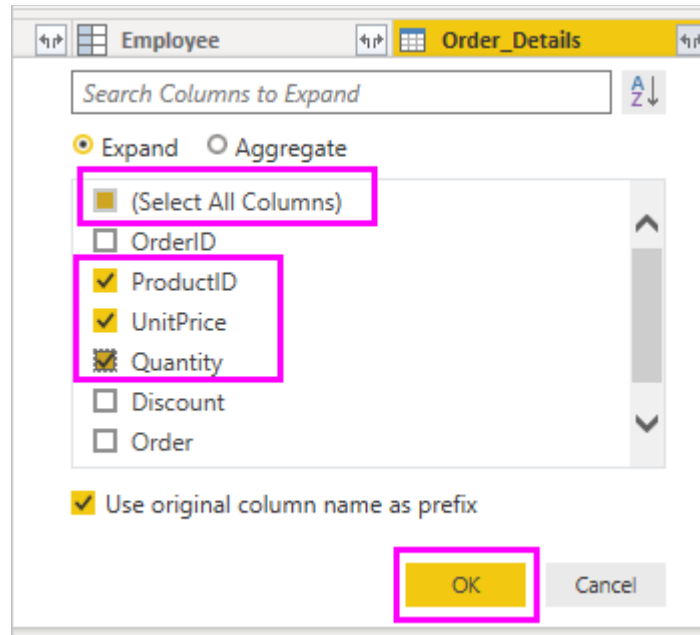
You can use table references to build queries when connecting to data sources with multiple tables, such as the Northwind OData feed. The **Orders** table contains references to several related tables from the Northwind dataset. You can use the expand operation to add the **ProductID**, **UnitPrice**, and **Quantity** columns from the related **Order_Details** table into the **Orders** table. Let's see how to do that.

1. Scroll to the right in the **Orders** table until you see the **Order_Details** column. It contains references to other tables but not data. These other tables are from the Northwind dataset, and they were "linked" with the Orders table (we will see later that these links are called relationships). For this reason, when you imported the Orders table, those links were kept and they are shown as "Table" in the image below.



The screenshot shows the Power Query Editor interface. The main area displays a table with columns: ipCountry, Customer, Employee, and Order_Details. The 'Order_Details' column is highlighted with a pink box, showing values like 'Table', 'Table', 'Table', etc. The 'Query Settings' pane on the right shows the 'Navigation' step.

2. To import part of those tables, select the **Expand** icon () in the **Order_Details** column header.
3. In the drop-down menu:
 1. Select **(Select All Columns)** to clear all columns.
 2. Select **ProductID**, **UnitPrice**, and **Quantity**, and then select **OK**.



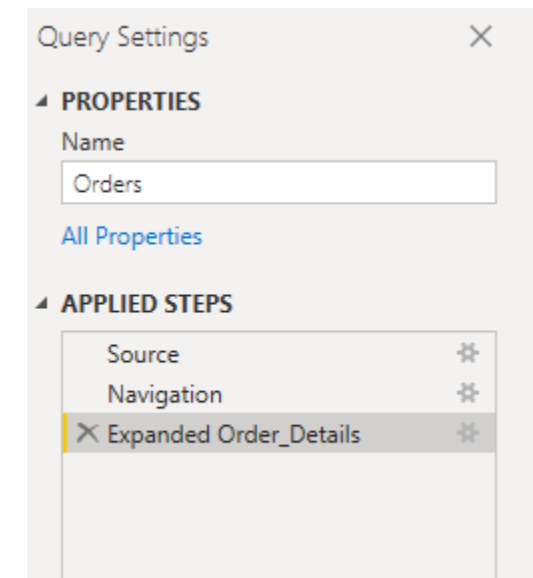
After you expand the **Order_Details** table, three new nested table columns replace the **Order_Details** column. There are new rows in the table for each order's added data. With this step, we have imported these three columns from the Order_Details Table to the Orders table.

fx = Table.ExpandTableColumn(Orders_table, "Order_Details", {"ProductID",

	1.2_3 Order_Details.ProductID	1.2 Order_Details.UnitPrice	1.2 Order_Details.Quantity	
1	11	14	12	Re
2	42	9.8	10	Re
3	72	34.8	5	Re
4	14	18.6	9	Re
5	51	42.4	40	Re
6	41	7.7	10	Re
7	51	42.4	35	Re
8	65	16.8	15	Re
9	22	16.8	6	Re
10	57	15.6	15	Re
11	65	16.8	20	Re
12	20	64.8	40	Re

A quick note: The Applied Steps pane

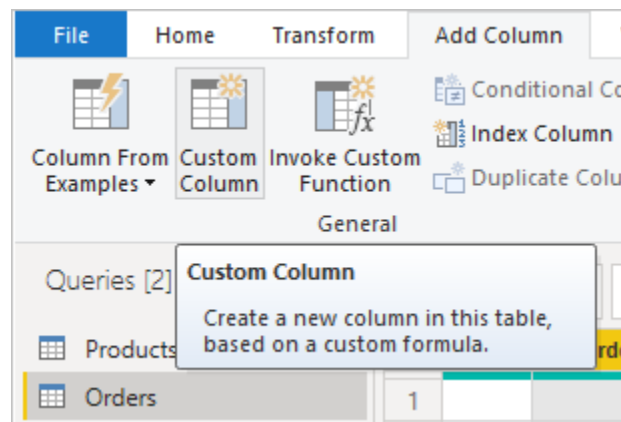
When working with Power BI, you might want to un-do a step, because you did something wrong or simply because you changed your mind. Remember that the list of performed tasks is in the Applied Step pane, on the right of the Query Editor. If you want to un-do a step, simply click on the cross next to the step and you will un-do that step.



Create a custom calculated column

In Power Query Editor, you can create calculations and custom fields to enrich your data. You will create a custom column that multiplies the unit price by item quantity to calculate the total price for each order's line item.

1. In the Power Query Editor's **Add Column** ribbon tab, select **Custom Column**.



2. In the **Custom Column** dialog box, type **LineTotal** in the **New column name** field. Line total will be the name of the new column.
3. In the **Custom column formula** field after the =, enter **[Order_Details.UnitPrice] * [Order_Details.Quantity]**. You can also select the field names from the **Available columns** scroll box and double click on them, instead of typing them.
4. Select **OK**.

Custom Column

Add a column that is computed from the other columns.

New column name

LineTotal

Custom column formula ⓘ

= [Order_Details.UnitPrice] * [Order_Details.Quantity]

[Learn about Power BI Desktop formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Available columns

- ShipPostalCode
- ShipCountry
- Customer
- Employee
- Order_Details.ProductID
- Order_Details.UnitPrice
- Order_Details.Quantity
- Shipper

<< Insert

The new **LineTotal** field appears as the last column in the **Orders** table (see next page).

Home Transform Add Column View Help

Custom Column Invoke Custom Function Duplicate Column General

Format ABC 123 Extract Parse From Text

Statistics Standard Scientific From Number

Trigonometry Rounding Information From Date & Time

Date Time Duration

[2]

Products

ers

Order_Details.ProductID 1.2 Order_Details.UnitPrice 1.2 Order_Details.Quantity Shipper Line Total

	Order_Details.ProductID	1.2 Order_Details.UnitPrice	1.2 Order_Details.Quantity	Shipper	Line Total
1	11	14	12	Record	168
2	42	9.8	10	Record	98
3	72	34.8	5	Record	174
4	14	18.6	9	Record	167.4
5	51	42.4	40	Record	1696
6	41	7.7	10	Record	77
7	51	42.4	35	Record	1484
8	65	16.8	15	Record	252
9	22	16.8	6	Record	100.8
10	57	15.6	15	Record	234
11	65	16.8	20	Record	336
12	20	64.8	40	Record	2592
13	33	2	25	Record	50
14	60	27.2	40	Record	1088
15	31	10	20	Record	200
16	39	14.4	42	Record	604.8
17	49	16	40	Record	640
18	24	3.6	15	Record	54
19	55	19.2	21	Record	403.2
20	74	8	21	Record	168
21	2	15.2	20	Record	304

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

Source

Navigation

Expanded Order_Details

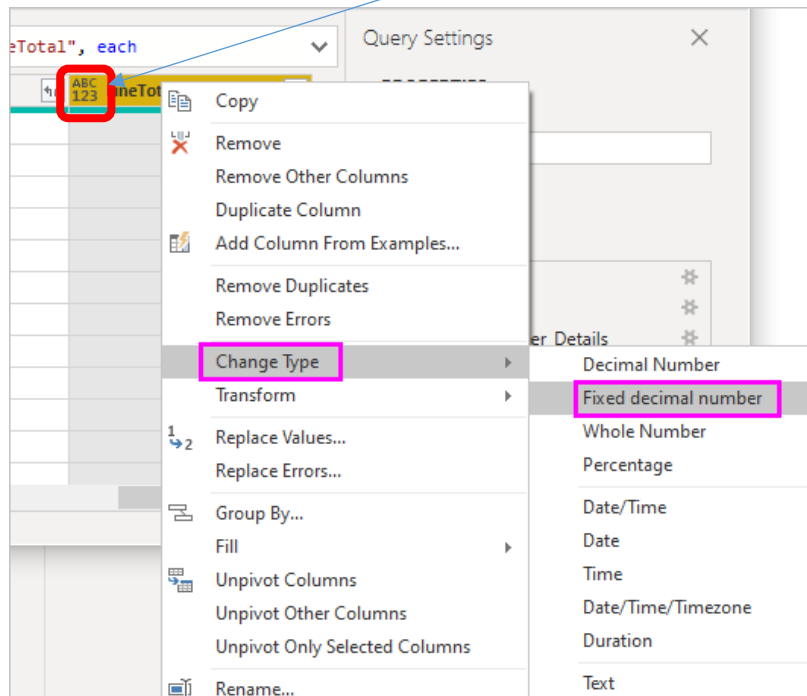
Added Custom

= Table.AddColumn("#Expanded Order_Details", "Line Total", each [Order_Details.UnitPrice]*

Set the new field's data type

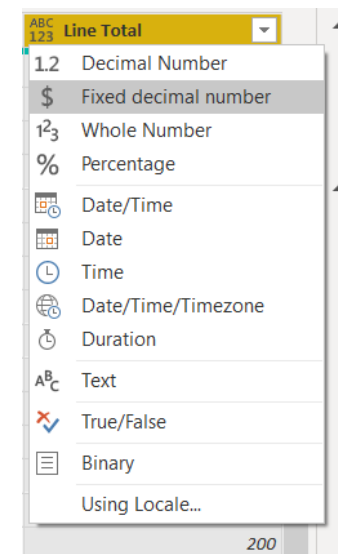
When Power Query Editor connects to data, it makes a best guess as to each field's data type for display purposes. A header icon indicates each field's assigned data type.

Your new **LineTotal** column has an **Any** data type (the symbol next to the column header), but it has currency values. To assign a data type, right-click the **LineTotal** column header, select **Change Type** from the drop-down menu, and then select **Fixed decimal number**.



Note

You can also select the **LineTotal** column, then click on the Data Type symbol and select Fixed decimal number as shown below.



Clean up the orders columns

To make your model easier to work with in reports, you can delete, rename, and reorder some columns.

Your report is going to use the following columns:

- **OrderDate**
- **ShipCity**
- **ShipCountry**
- **Order_Details.ProductID**
- **Order_Details.UnitPrice**
- **Order_Details.Quantity**
- **LineTotal**

Select these columns by keep pressing Ctrl in your keyboard and click on each of them. Then, right click on one of the columns' headers and select use **Remove Other Columns** to remove all the columns that you haven't selected. Otherwise, you can select the non-listed columns, right-click on one of them, and select **Remove Columns**.

You can rename the columns prefixed with "**Order_Details.**" to make them easier to read:

1. Double-click or tap and hold each column header, or right-click the column header, and select **Rename** from the drop-down menu.
2. Delete the **Order_Details.** prefix from each name. Hence, the final names will be: **ProductID, UnitPrice and Quantity.**

Finally, to make the **LineTotal** column easier to access, drag and drop it to the left, just to the right of the **ShipCountry** column.

	OrderDate	A ^B _C ShipCity	A ^B _C ShipCountry	\$ LineTotal	1 ² ₃ ProductID	1.2 UnitPrice	1.2 Quantity
1	04/07/1996 00:00:00	Reims	France	168.00	11	14	12
2	04/07/1996 00:00:00	Reims	France	98.00	42	9.8	10
3	04/07/1996 00:00:00	Reims	France	174.00	72	34.8	5
4	05/07/1996 00:00:00	Münster	Germany	167.40	14	18.6	9
5	05/07/1996 00:00:00	Münster	Germany	1,696.00	51	42.4	40
6	08/07/1996 00:00:00	Rio de Janeiro	Brazil	77.00	41	7.7	10
7	08/07/1996 00:00:00	Rio de Janeiro	Brazil	1,484.00	51	42.4	35
8	08/07/1996 00:00:00	Rio de Janeiro	Brazil	252.00	65	16.8	15
9	08/07/1996 00:00:00	Lyon	France	100.80	22	16.8	6
10	08/07/1996 00:00:00	Lyon	France	234.00	57	15.6	15
11	08/07/1996 00:00:00	Lyon	France	336.00	65	16.8	20
12	09/07/1996 00:00:00	Charleroi	Belgium	2,592.00	20	64.8	40

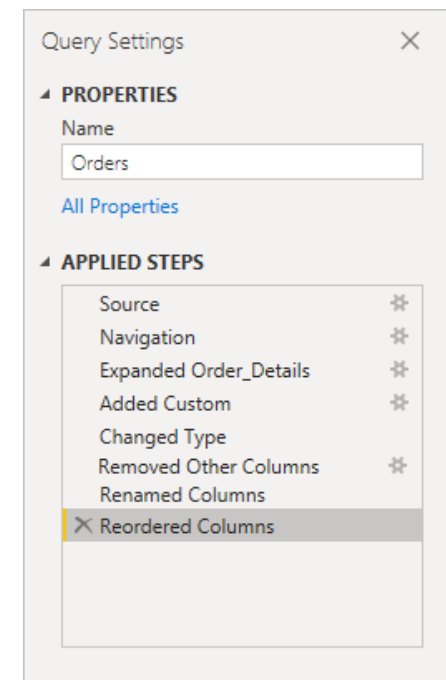
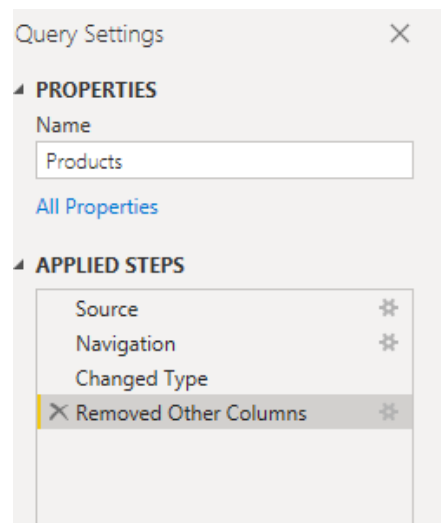
profiling based on top 1000 rows

Review the query steps

If during any of the steps above, you did any mistake, do not worry. Your Power Query Editor actions to shape and transform data are recorded. Each action appears on the right in the **Query Settings** pane under **Applied Steps**. You can step back through the **Applied Steps** to review your steps, and edit, delete, or rearrange them if necessary.

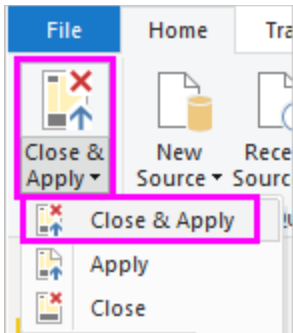
If you perform the wrong step and you want to un-do it, simply click on the cross next to the corresponding step.

Select each of your queries in the **Queries** list on the right side of Power Query Editor, and review the **Applied Steps** in **Query Settings**. After applying the previous data transformations, the **Applied Steps** for your two queries should look like this:



Import the transformed queries

When you're satisfied with your transformed data and ready to import it into Power BI Desktop **Report** view, select **Close & Apply** > **Close & Apply** in the **Home** ribbon tab's **Close** group.



Once the data is loaded, the queries appear in the **Fields** list in the Power BI Desktop **Report** view. Save your file by clicking on File, Save and save it as Task 2. Well done you have reached the end of this tutorial! Your data is now ready to be used for creating beautiful reports. We will see how to do this in our next lessons.

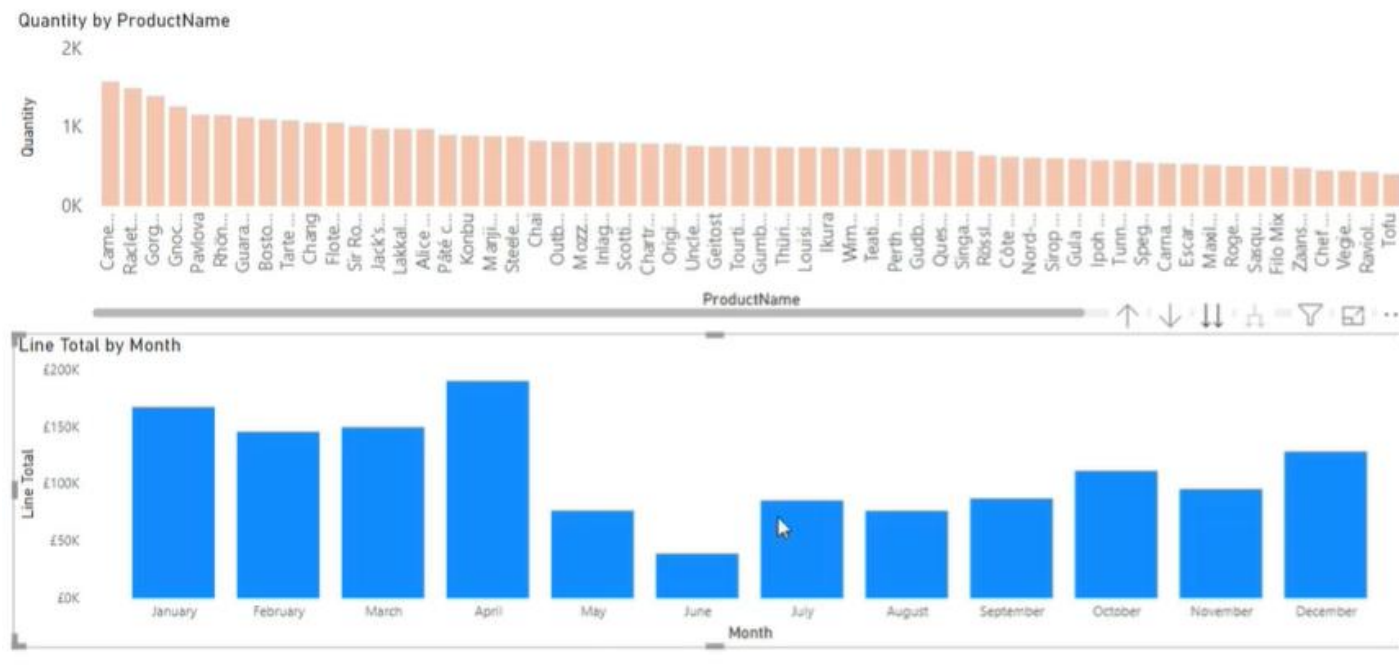
Your turn! Create the final Dashboard

Now, that you have performed some pre-processing steps it's time to create a dashboard.

Start thinking about the business questions that you might be able to answer when working with this dataset. Once you have identified two or three Business Questions, start working on your dashboard.

Don't forget to share your final work (and the Business Questions you're trying to answer) on Padlet!

Can you generate the charts below?



Can you address the Business Questions below with your dashboard?

- Which products are the top-10 best-selling products? How many products can you see? Are they actually 10? Why?
- Where are most of our global sales made of?
- Which quarters and months do we need to target our attention to sustain growth throughout the year?
- Which are the top-5 countries bringing the highest revenue?

What about the following ones?



Remember that you have plenty of charts you can add and explore using the Get More Visuals option.

